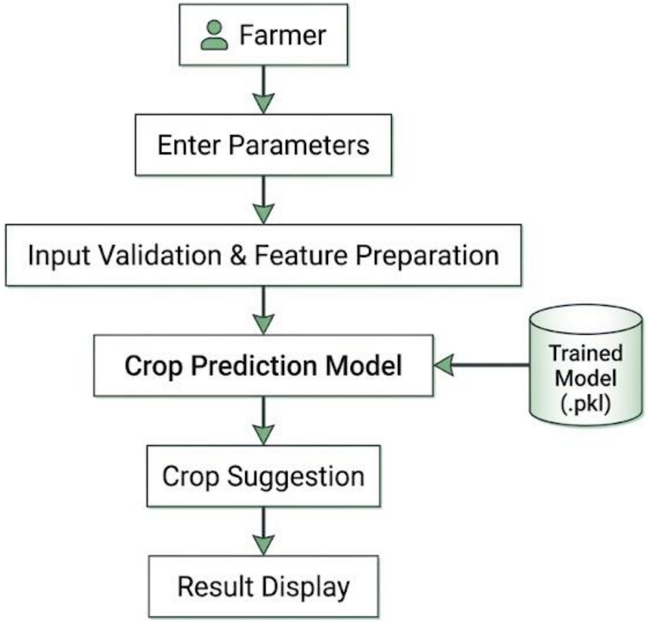
Project Design Phase-II
Data Flow Diagram & User Stories

Date	29 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.



User Stories

User Type	Functional Requirement	User Story No.	User Story	Acceptance Criteria	Priority	Release
Farmer	Soil Data	US-1	As a farmer, I want to enter Nitrogen (N), Phosphorus (P), and Potassium (K) values so that the system can analyze my soil nutrients.	N, P, and K values accepted successfully.	High	Sprint 1
Farmer	Soil Data	US-2	As a farmer, I want to enter the soil pH value so that the system can evaluate soil suitability.	pH value accepted successfully.	High	Sprint 1
Farmer	Environmental Data	US-3	As a farmer, I want to enter temperature, humidity, and rainfall values so that environmental conditions can be analyzed.	Environmental parameters accepted successfully.	High	Sprint 1
Farmer	Input Validation	US-4	As a farmer, I want the system to validate my soil and environmental inputs before generating recommendations.	Invalid or incomplete inputs are detected and appropriate messages are displayed.	High	Sprint 1
Farmer	Crop Recommendation	US-5	As a farmer, I want the system to recommend the most suitable crop based on my soil and environmental conditions.	Recommended crop is generated successfully.	High	Sprint 2
Farmer	Prediction Results	US-6	As a farmer, I want to view the recommended crop on the results page after submitting my inputs.	Crop recommendation is displayed successfully.	High	Sprint 2
Farmer	Machine Learning Prediction	US-7	As a farmer, I want the system to use a trained machine learning model for accurate crop prediction.	Prediction is generated using the trained model.	Medium	Sprint 2
Farmer	User Interface	US-8	As a farmer, I want a simple and easy-to-use interface for entering my farm data and viewing recommendations.	User can navigate the application and obtain recommendations without difficulty.	Medium	Sprint 3